

WDC - Western Digital | Full Deep Dive

September 21, 2026 | Framework 6.0.2 | NOT HELD | BOTTLENECK - qualified nearline HDD capacity | High Conviction

Analytical Base: 46/60 (77%). Valuation: 20x forward four-quarter adjusted operating profit.

Snapshot: USD 448.17 per share on September 21, 2026 close; 360.541 million shares, implying USD 161,583.63 million equity value. Source [WDC-6]. Price is a dated header snapshot, not a valuation input in the body.

Active horizon Q3 2027 ceiling: USD 727.67/share. Q1 2028: USD 842.40/share.

All financial model amounts are USD millions, shares in millions, unless stated. Calendar buckets map to the nearest fiscal quarter: Q3 2026 = FQ1 FY2027, Q4 = FQ2, Q1 2027 = FQ3 and Q2 = FQ4. Fiscal year ends around June; July 3 actuals belong to the June/Q2 bucket. The fiscal mapping is shown for every modeled quarter.

1. Thesis, bottleneck and expanded TAM

Western Digital is a standalone HDD company after the February 2025 SanDisk separation. The opportunity is capacity storage for cloud and AI: data accumulates as compute is reused, and HDDs provide an economical persistent tier. The company sells heads/media engineering, manufacturing yield, qualified reliability and capacity per drive. It is not a NAND producer in this model. [WDC-2]

Scarcity rests on a small qualified supplier base and long manufacturing and validation cycles. Higher platter density expands exabytes without a proportional increase in finished-drive units. Western Digital can monetize its current ePMR/UltraSMR platform while preparing HAMR, limiting the need for an abrupt migration. This dual path is an execution choice, not proof of a permanent monopoly.

The next opportunities are persistent inference context, video, sovereign/enterprise clouds and physical-AI data. A WD-sponsored IDC survey found 61% of surveyed organizations had AI-related data growth of at least 25% in the past year. This is supporting demand evidence, not an industry revenue TAM or a booking. The model treats the installed-data effect within a capped exabyte demand bridge. [WDC-8]

Seagate is the primary scaled peer and already has commercial HAMR experience; Toshiba is the other major HDD producer. SSD/NAND suppliers compete in high-performance tiers and at the storage boundary. HDD supply discipline can break if density or capacity expands faster than demand, even without a new entrant. No microcap direct substitute for a fully qualified nearline HDD platform was identified. UltraSMR is a recording/data-placement approach that may coexist with ePMR or HAMR; their market shares cannot be added as separate TAMs. [WDC-2, WDC-3]

TAM expansion and S-curve

Phase	Economic exposure	Model treatment
2026-27: present storage wallet	Cloud remains the dominant reported end market; 40TB ePMR enters volume	A capacity and pricing step inside the existing HDD revenue pool.
2027-28: HAMR and performance	44TB qualification, then higher density and high-bandwidth functions	Probability-weighted HAMR shipments; all replace part of older capacity.
2029-30+: compounding retained data	Wider enterprise, physical AI and larger data lakes	Retention demand index; no separate robotics or all-flash displacement TAM added.

AH-12 / Plan-B demand bridge: customer identities and supplier wallet shares are confidential. Amazon, Microsoft, Alphabet and Meta form a representative spending basket, not a claim that each is a disclosed Western Digital customer. Their latest CY2026 capex figures are \$220B, \$175B, \$200B midpoint and \$137.5B midpoint respectively, totaling \$732.5B [D1-D4]. Definitions differ: Amazon is cash capex, Meta includes finance-lease principal, and Microsoft changed lease classification; the basket is a demand index, not HDD TAM. Memory inflation is not extra physical demand. Calibrate one effective supplier coefficient against the latest actual quarter: $\$3,335\text{M} / (\$732,500\text{M} / 4) = 1.82116\%$. Applying it to the quarterly run-rate basket reproduces \$3,335M exactly (0.0% calibration error). This coefficient embeds non-basket cloud/enterprise business; individual customer shares are NOT separately knowable. Calibration of one point does not establish forecast accuracy. Each customer

is extended at the same analyst-assumed capex growth: +25% in 2027, +20% in 2028, +14% in 2029 and +10% in 2030. Thus basket capex is \$915.625B, \$1,098.750B, \$1,252.575B and \$1,377.833B. Quarterly allocation is one quarter of each annual total, a normalization rather than a forecast of actual quarterly cash spending. Demand EB = 207.9 anchor EB x (basket capex / \$732.5B) x retained-data intensity. Intensity equals 1.15 / 1.20 in Q3/Q4 2026, 1.25 in 2027-28, 1.30 in 2029 and 1.45 in 2030; these LOW-confidence analyst indices represent accumulating installed-base data. Potential EB = MIN(demand EB, manufacturing capacity EB). Apply the separate HAMR qualification reserve to determine shipped EB. Revenue then equals shipped EB x modeled USD/TB, with no second capex multiplier. Capacity, demand, mix and ASP are separately visible below. This is a calibrated proxy, not a company-disclosed customer model. The company reports 231 total EB in the latest quarter, not a precise cloud-only EB figure in the source set. The model allocates 90% to cloud (207.9 EB) as a LOW-confidence physical proxy; \$3,335M cloud revenue is independently anchored to the reported \$3,747M total less \$225M client and \$187M consumer. Changing that EB split changes implied USD/TB, not the calibrated actual cloud revenue.

2. Latest evidence and financial anchor

Latest two earnings calls reviewed: April 30 and August 5, 2026, including management answers. April established strong cloud demand, four HAMR qualifications, long-duration customer planning and continuing cost-per-TB improvements. Its then-current June guidance is superseded by the July 3 actuals in the August release. [WDC-1, WDC-3, WDC-4]

August management placed 40TB ePMR into volume ramp at two customers, targeted at least half of nearline exabytes on that platform by FQ3 FY2027, and expected UltraSMR to reach about 60% by fiscal year-end. It described 44TB HAMR timing in H1 2027 and high-bandwidth samples at five customers. Long-range LTA discussions into 2031 are not the same as fully priced, binding orders for that entire period. [WDC-3]

The September 9 update sharpened timing: small HAMR shipments in Q1 2027, more in Q2, then a broader H2 ramp. It also identified DRAM/NAND input-cost inflation and indicated capex could run slightly above the long-term 4-6% range over the next couple of years. Those costs matter even though the company benefits indirectly from high NAND prices. [WDC-5]

Period	Status	Revenue	Gross profit	Operating profit	Source
Q1 2026 / FQ3 FY26	Actual adjusted	3,337.00	1,684.00	1,287.00	WDC-1
Q2 2026 / FQ4 FY26	Actual GAAP	3,747.00	2,028.00	1,563.00	WDC-1
Q2 2026 / FQ4 FY26	Actual adjusted	3,747.00	2,037.00	1,655.00	WDC-1

Numerical anchor: \$3,747M sales = \$3,335M cloud + \$225M client + \$187M consumer. Adjusted GP is \$2,037M and opex \$382M, producing \$1,655M OP. GAAP GP is \$2,028M and opex \$465M; +\$9M GP and -\$83M opex reconcile the \$92M adjusted OP difference. For an actual-quarter product-margin proxy, assume 52% client/consumer GM: \$412M x 52% = \$214.24M GP, leaving \$1,822.76M cloud GP / \$3,335M = 54.66%. This fits management description that both categories are now in the 50s. It does not claim the company disclosed those exact segment margins. [WDC-1, WDC-3]

FQ1 FY2027 revenue guidance is \$4,100M +/- \$100M, adjusted GM 55-56% and opex \$390-400M. The model uses the midpoints: \$4,100M sales, \$2,275.5M GP and \$395M opex, yielding \$1,880.5M OP (45.87%). The revenue split is \$3,650M cloud and \$450M client/consumer. At 223 modeled cloud EB, realized cloud price is \$16.368/TB. Current guided adjusted EPS of \$4.00 +/- \$0.15, 17% tax and 388M diluted shares provide a consistency check. They are not substituted for operating profit. [WDC-1, WDC-3]

Balance sheet and capital allocation

July 3 cash was \$1,579M against roughly \$1.1B debt. The remaining 1.7M SanDisk shares were monetized in the quarter; the model adds no residual SanDisk stake or flash earnings. On September 14, WDC called the remaining \$109.505M of 3% convertible notes for November 16 redemption; that is a future event, not already completed debt removal. [WDC-1, WDC-2, WDC-7] The cash-flow cross-check allows capex at 7% of revenue in 2027, 6.5% in 2028, 6% in 2029 and 5.5% in 2030, reflecting September commentary. These are analyst assumptions; normal depreciation remains in vector costs. Buybacks are not converted into an assumed falling denominator.

3. Base score and evidence ledger

Category	Score	Evidence
Core Thesis	17	Scaled, qualified head/media platform and few alternatives; Seagate lead prevents monopoly credit
Cycle Stage	9	40TB ramp is underway; HAMR remains a qualification-to-volume transition
TAM Expansion	8	Persistent AI data and wider cloud deployment grow the storage wallet
Catalysts	5	40TB adoption, H1 2027 HAMR start and later performance features
Macro	9	Cloud capex and data retention are strong; construction/power limits affect timing
Alpha	2	Broadly followed; edge comes from separating NAND, HDD and qualification economics
Risks	-4	Four independent risk classes below
Total	46/60 = 77%	High Conviction

46/60 = 77% (High Conviction). The business has a durable infrastructure role, but the company itself is not unavoidable: other qualified HDD suppliers exist. Four distinct risk deductions are applied once. Base is independent of market price, buybacks and the chosen multiple.

4. Quarterly operating model

Four revenue vectors: HAMR cloud, 40TB-class ePMR cloud, other cloud drives, and client/consumer. Potential cloud EB = MIN(capex-derived demand, physical capacity). Allocate potential EB between planned HAMR, higher-density ePMR and residual legacy. Multiply only the planned HAMR units by their explicit qualification/delivery probability; missed HAMR units are not automatically replaced. Normalize a 4% HAMR and 2% advanced-ePMR price/TB premium so shipped cloud EB reproduce the displayed blended ASP. These premiums are analyst assumptions, not disclosed product pricing. Each vector has a separate GM and GP; R&D/SG&A use explicit revenue percentages. UltraSMR is embedded in product economics, never added as another shipment pool.

Revenue vector	Driver and source	Confidence
HAMR cloud drives	WDC-3/5: four qualifications and H1 2027 start; planned EB x conversion factor below x normalized ASP.	MEDIUM timing; LOW mix and product GM
40TB-class ePMR cloud drives	WDC-3: current volume ramp and FQ3 FY27 >50% mix ambition; modeled mix declines as HAMR replaces it.	MEDIUM 2026-27; LOW later
Other cloud drives	WDC-1/3: remaining calibrated cloud pool after newer products; explicit cannibalization.	MEDIUM near term; LOW later
Client and consumer HDDs	WDC-1/3/5: \$412M actual; stronger NAND supports pricing but also raises component costs.	MEDIUM Q3 2026; LOW later

Raw cloud capacity rises from 223 EB in Q3 2026 to 318 EB in Q4 2027 and 584 EB in Q4 2030; demand and HAMR qualification can restrict realized shipments. This is a density/automation supply proxy, not announced factory guidance. Cloud ASP rises 2.5%, 2.5%, 2%, 2%, 1.5% sequentially through Q4 2027, then is flat in 2028 and eases 1% annually in 2029 and 2% in 2030. HAMR GM ramps to 70%, advanced-ePMR peaks at 64.5% before mix runoff, other cloud settles at 52.5%, and client/consumer at 57%. Later margins are analyst estimates with LOW confidence. Opex rises from \$395M to \$516M quarterly; R&D/SG&A use a 67%/33% allocation informed by reported expense mix, not management segment forecasts. No stock-compensation cash add-back is confused with a lower share count.

Physical and commercial drivers

Calendar / fiscal	Demand EB	Capacity EB	Ship EB	USD/TB	HAMR mix (EB)	Main driver / confidence
Q3 2026 / FQ1 FY2027	239.1	223.0	223.0	16.368	0.0%	Guide / MEDIUM
Q4 2026 / FQ2 FY2027	249.5	241.0	241.0	16.777	0.0%	Ramp / MEDIUM
Q1 2027 / FQ3 FY2027	324.8	263.0	260.4	17.196	4.0%	Ramp / MEDIUM
Q2 2027 / FQ4 FY2027	324.8	285.0	279.9	17.540	10.4%	Ramp / MEDIUM
Q3 2027 / FQ1 FY2028	324.8	301.0	294.4	17.891	20.2%	Ramp / MEDIUM
Q4 2027 / FQ2 FY2028	324.8	318.0	312.4	18.159	33.8%	Ramp / MEDIUM
Q1 2028 / FQ3 FY2028	389.8	335.0	326.6	18.159	48.7%	Capacity / LOW
Q2 2028 / FQ4 FY2028	389.8	354.0	342.5	18.159	63.8%	Capacity / LOW
Q3 2028 / FQ1 FY2029	389.8	378.0	363.8	18.159	74.0%	Capacity / LOW
Q4 2028 / FQ2 FY2029	389.8	403.0	373.2	18.159	84.3%	Capacity / LOW
Q1 2029 / FQ3 FY2029	462.2	424.0	404.9	18.114	89.5%	Capacity / LOW
Q2 2029 / FQ4 FY2029	462.2	446.0	425.5	18.068	91.6%	Capacity / LOW
Q3 2029 / FQ1 FY2030	462.2	468.0	440.4	18.023	93.7%	Capacity / LOW
Q4 2029 / FQ2 FY2030	462.2	490.0	440.2	17.978	94.8%	Capacity / LOW
Q1 2030 / FQ3 FY2030	567.0	512.0	487.6	17.887	95.3%	Capacity / LOW
Q2 2030 / FQ4 FY2030	567.0	536.0	510.3	17.797	95.8%	Capacity / LOW
Q3 2030 / FQ1 FY2031	567.0	560.0	533.0	17.707	96.3%	Capacity / LOW
Q4 2030 / FQ2 FY2031	567.0	584.0	539.5	17.618	96.8%	Capacity / LOW

Qualification and product allocation

The HAMR factor is 80% in Q1 2027, 85% in Q2, 90% in Q3 and 95% from Q4 onward; this is an analyst qualification-and-delivery reserve applied once to the planned units. Zero HAMR revenue is included in 2026. The stronger weight over time follows the management-stated initial-shipment/ramp sequence, not a claim that future milestones have already occurred. The forward product mix is low-confidence. Existing ePMR/legacy production uses 100% of its planned share. If documented qualification slips, reduce the HAMR factor and its associated exabytes; do not haircut the full company ceiling again. 40TB-class ePMR share is 50% of pre-reserve cloud EB in Q1 2027, consistent with the platform milestone. The high-bandwidth roadmap is not a separate revenue pool.

Quarter	Planned HAMR mix	HAMR factor	40TB ePMR mix	HAMR ship EB	ePMR ship EB	Other cloud EB
Q3 2026	0.0%	100%	25.0%	0.00	55.75	167.25
Q4 2026	0.0%	100%	40.0%	0.00	96.40	144.60
Q1 2027	5.0%	80%	50.0%	10.52	131.50	118.35
Q2 2027	12.0%	85%	60.0%	29.07	171.00	79.80
Q3 2027	22.0%	90%	55.0%	59.60	165.55	69.23
Q4 2027	35.0%	95%	45.0%	105.73	143.10	63.60
Q1 2028	50.0%	95%	35.0%	159.12	117.25	50.25
Q2 2028	65.0%	95%	25.0%	218.59	88.50	35.40
Q3 2028	75.0%	95%	18.0%	269.32	68.04	26.46
Q4 2028	85.0%	95%	12.0%	314.77	46.78	11.69
Q1 2029	90.0%	95%	8.0%	362.52	33.92	8.48
Q2 2029	92.0%	95%	5.0%	389.80	22.30	13.38
Q3 2029	94.0%	95%	3.0%	412.71	13.86	13.86
Q4 2029	95.0%	95%	2.0%	417.10	9.24	13.86
Q1 2030	95.5%	95%	1.0%	464.51	5.12	17.92
Q2 2030	96.0%	95%	0.5%	488.83	2.68	18.76
Q3 2030	96.5%	95%	0.2%	513.38	1.12	18.48
Q4 2030	97.0%	95%	0.0%	522.52	0.00	17.01

Vector revenue, gross margin and gross profit

For each row: vector gross profit = vector revenue x its own gross margin. The following tables expose the inputs; they are analyst model assumptions unless explicitly identified as reported or guided.

HAMR cloud drives

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	0.00	58.00%	0.00
Q4 2026	0.00	58.00%	0.00
Q1 2027	185.96	59.00%	109.72
Q2 2027	521.75	61.00%	318.27
Q3 2027	1,087.88	63.50%	690.80
Q4 2027	1,952.57	65.50%	1,278.93
Q1 2028	2,927.15	66.50%	1,946.55
Q2 2028	4,005.39	67.50%	2,703.64
Q3 2028	4,922.26	68.50%	3,371.75
Q4 2028	5,736.85	69.50%	3,987.11
Q1 2029	6,582.54	70.00%	4,607.78
Q2 2029	7,058.79	70.00%	4,941.15
Q3 2029	7,451.84	70.00%	5,216.29
Q4 2029	7,510.70	70.00%	5,257.49
Q1 2030	8,322.29	70.00%	5,825.60
Q2 2030	8,713.01	70.00%	6,099.11
Q3 2030	9,103.18	70.00%	6,372.22
Q4 2030	9,217.16	70.00%	6,452.01

40TB-class ePMR cloud drives

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	926.12	58.00%	537.15
Q4 2026	1,636.55	59.00%	965.56
Q1 2027	2,279.83	60.50%	1,379.30
Q2 2027	3,010.08	62.00%	1,866.25
Q3 2027	2,963.77	63.00%	1,867.17
Q4 2027	2,591.76	64.00%	1,658.73
Q1 2028	2,115.37	64.50%	1,364.41
Q2 2028	1,590.43	64.50%	1,025.83
Q3 2028	1,219.60	64.00%	780.55
Q4 2028	836.14	64.00%	535.13
Q1 2029	604.07	63.50%	383.58
Q2 2029	396.05	63.50%	251.49
Q3 2029	245.53	63.00%	154.68
Q4 2029	163.24	63.00%	102.84
Q1 2030	89.97	63.00%	56.68
Q2 2030	46.85	63.00%	29.52
Q3 2030	19.48	63.00%	12.27
Q4 2030	0.00	63.00%	0.00

Other cloud drives

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	2,723.88	54.90%	1,495.35
Q4 2026	2,406.69	55.50%	1,335.71
Q1 2027	2,011.62	56.00%	1,126.50
Q2 2027	1,377.16	56.00%	771.21
Q3 2027	1,215.09	55.50%	674.38
Q4 2027	1,129.31	55.00%	621.12
Q1 2028	888.81	54.50%	484.40
Q2 2028	623.70	54.00%	336.80
Q3 2028	464.99	53.50%	248.77
Q4 2028	204.94	53.00%	108.62
Q1 2029	148.06	52.50%	77.73
Q2 2029	232.97	52.50%	122.31
Q3 2029	240.71	52.50%	126.37
Q4 2029	240.06	52.50%	126.03
Q1 2030	308.71	52.50%	162.07
Q2 2030	321.52	52.50%	168.80
Q3 2030	315.08	52.50%	165.42
Q4 2030	288.53	52.50%	151.48

Client and consumer HDDs

Quarter	Revenue	Gross margin	Gross profit
Q3 2026	450.00	54.00%	243.00
Q4 2026	480.00	55.00%	264.00
Q1 2027	475.00	55.50%	263.62
Q2 2027	490.00	56.00%	274.40
Q3 2027	500.00	56.50%	282.50
Q4 2027	515.00	57.00%	293.55
Q1 2028	505.00	57.00%	287.85
Q2 2028	515.00	57.50%	296.12
Q3 2028	525.00	57.50%	301.88
Q4 2028	540.00	58.00%	313.20
Q1 2029	525.00	57.50%	301.88
Q2 2029	535.00	57.50%	307.62
Q3 2029	545.00	57.00%	310.65
Q4 2029	560.00	57.00%	319.20
Q1 2030	540.00	57.00%	307.80
Q2 2030	550.00	57.00%	313.50
Q3 2030	560.00	57.00%	319.20
Q4 2030	575.00	57.00%	327.75

Combined quarterly earnings

Quarter	Status	Revenue	Gross profit	R&D	SG&A	Other opex	OP	OP margin
Q3 2026	Model	4,100.00	2,275.50	264.65	130.35	0.00	1,880.50	45.87%
Q4 2026	Model	4,523.23	2,565.27	268.00	132.00	0.00	2,165.27	47.87%
Q1 2027	Model	4,952.41	2,879.14	271.35	133.65	0.00	2,474.14	49.96%
Q2 2027	Model	5,398.99	3,230.13	274.70	135.30	0.00	2,820.13	52.23%
Q3 2027	Model	5,766.73	3,514.85	278.05	136.95	0.00	3,099.85	53.75%
Q4 2027	Model	6,188.64	3,852.33	281.40	138.60	0.00	3,432.33	55.46%
Q1 2028	Model	6,436.32	4,083.22	286.09	140.91	0.00	3,656.22	56.81%
Q2 2028	Model	6,734.51	4,362.38	290.78	143.22	0.00	3,928.38	58.33%
Q3 2028	Model	7,131.85	4,702.94	295.47	145.53	0.00	4,261.94	59.76%
Q4 2028	Model	7,317.92	4,944.05	300.16	147.84	0.00	4,496.05	61.44%
Q1 2029	Model	7,859.66	5,370.96	305.52	150.48	0.00	4,914.96	62.53%
Q2 2029	Model	8,222.82	5,622.58	310.88	153.12	0.00	5,158.58	62.73%
Q3 2029	Model	8,483.08	5,807.99	316.24	155.76	0.00	5,335.99	62.90%
Q4 2029	Model	8,474.00	5,805.57	321.60	158.40	0.00	5,325.57	62.85%
Q1 2030	Model	9,260.97	6,352.16	327.63	161.37	0.00	5,863.16	63.31%
Q2 2030	Model	9,631.39	6,610.92	333.66	164.34	0.00	6,112.92	63.47%
Q3 2030	Model	9,997.74	6,869.11	339.69	167.31	0.00	6,362.11	63.64%
Q4 2030	Model	10,080.69	6,931.24	345.72	170.28	0.00	6,415.24	63.64%

Quarterly drivers and confidence

Calendar / fiscal	Status	Revenue	OP margin	OP	Main driver	Confidence
Q3 2026 / FQ1 FY2027	Model	4,100.00	45.87%	1,880.50	40TB ePMR ramp	MEDIUM
Q4 2026 / FQ2 FY2027	Model	4,523.23	47.87%	2,165.27	Higher 40TB mix	MEDIUM
Q1 2027 / FQ3 FY2027	Model	4,952.41	49.96%	2,474.14	HAMR first shipments	MEDIUM
Q2 2027 / FQ4 FY2027	Model	5,398.99	52.23%	2,820.13	HAMR conversion	MEDIUM
Q3 2027 / FQ1 FY2028	Model	5,766.73	53.75%	3,099.85	HAMR volume ramp	MEDIUM
Q4 2027 / FQ2 FY2028	Model	6,188.64	55.46%	3,432.33	Wider acceptance	MEDIUM
Q1 2028 / FQ3 FY2028	Model	6,436.32	56.81%	3,656.22	Density migration	LOW
Q2 2028 / FQ4 FY2028	Model	6,734.51	58.33%	3,928.38	Broader qualification	LOW
Q3 2028 / FQ1 FY2029	Model	7,131.85	59.76%	4,261.94	HAMR-led supply	LOW
Q4 2028 / FQ2 FY2029	Model	7,317.92	61.44%	4,496.05	Demand / capacity limit	LOW
Q1 2029 / FQ3 FY2029	Model	7,859.66	62.53%	4,914.96	Density / retention	LOW
Q2 2029 / FQ4 FY2029	Model	8,222.82	62.73%	5,158.58	Density / retention	LOW
Q3 2029 / FQ1 FY2030	Model	8,483.08	62.90%	5,335.99	Density / retention	LOW
Q4 2029 / FQ2 FY2030	Model	8,474.00	62.85%	5,325.57	Density / retention	LOW
Q1 2030 / FQ3 FY2030	Model	9,260.97	63.31%	5,863.16	Retained-data growth	LOW
Q2 2030 / FQ4 FY2030	Model	9,631.39	63.47%	6,112.92	Density efficiency	LOW
Q3 2030 / FQ1 FY2031	Model	9,997.74	63.64%	6,362.11	Capacity execution	LOW
Q4 2030 / FQ2 FY2031	Model	10,080.69	63.64%	6,415.24	Demand / capacity limit	LOW

Annual trajectory cross-check

The calendar-bucket sums below are a capacity and demand cross-check; quarterly windows remain the live valuation method. Years beyond the latest guidance are model extensions, not company guidance.

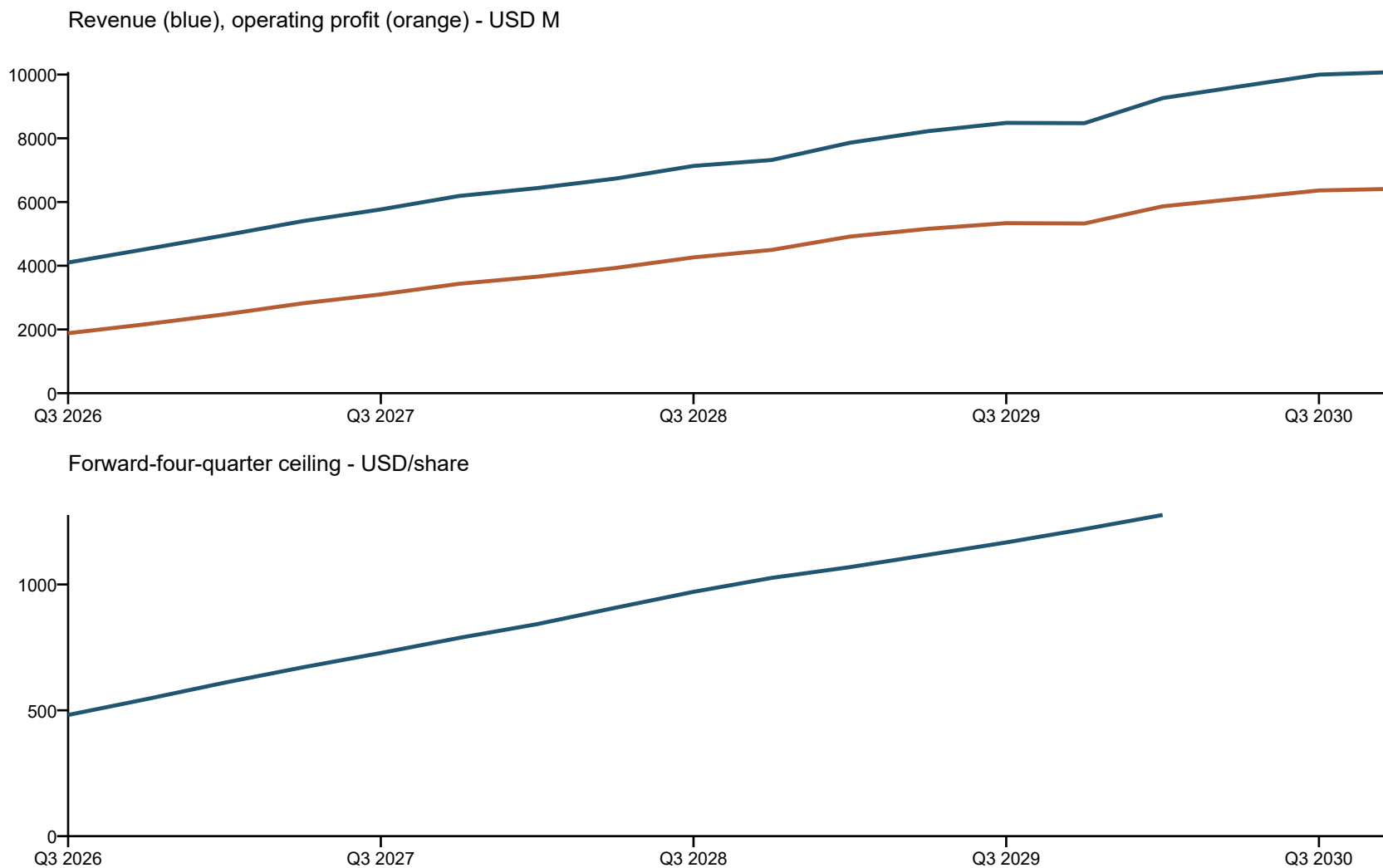
Year	Revenue	Operating profit	OP margin
2027	22,306.77	11,826.45	53.02%
2028	27,620.61	16,342.59	59.17%
2029	33,039.56	20,735.10	62.76%
2030	38,970.78	24,753.43	63.52%

FY2027 is the first four modeled quarters, separate from the calendar-year sums. Compare it with \$12,919M FY2026 revenue; WDC did not provide a full-year dollar guide in the latest call. CY2028 growth depends on both greater supply and a qualified HAMR ramp. Revenue is below the unconstrained demand opportunity where supply or qualification limits bind. The published technology roadmap is not multiplied into a second TAM. Long-dated spending, storage intensity and margins require recurring recalibration.

Fiscal-year cross-check

Fiscal year	Exact model window	Revenue	Operating profit
2027	Q3 2026 to Q2 2027	18,974.63	9,340.05
2028	Q3 2027 to Q2 2028	25,126.21	14,116.78
2029	Q3 2028 to Q2 2029	30,532.25	18,831.54
2030	Q3 2029 to Q2 2030	35,849.44	22,637.64

5. Forward four-quarter ceiling and derivations



Active valuation policy, selected by the portfolio owner on September 21, 2026: 20x forward four-quarter adjusted operating profit for both Seagate and Western Digital. This is a chosen operating-profit multiple under the framework convention, rather than an observed market P/E. The next 12-16 months carry the greatest decision weight, with uncertainty increasing further out. Longer-dated projections remain conditional scenarios. Product execution differences remain in each operating model and Base score.

The framework convention applies the operating-profit multiple directly to diluted equity shares. It is neither a net-income P/E nor an enterprise-value bridge. No separate net-cash addition, tax conversion, probability haircut or price-dependent ranking enters this ceiling. Any probability-weighted revenue is already included once in its operating vector.

Ceiling quarter	Exact earnings window	Forward-4Q OP	Multiple	Diluted shares	Ceiling/share
Q3 2026	Q3 2026 to Q2 2027	9,340.05	20x	388.000	481.45
Q4 2026	Q4 2026 to Q3 2027	10,559.40	20x	388.000	544.30
Q1 2027	Q1 2027 to Q4 2027	11,826.45	20x	388.000	609.61
Q2 2027	Q2 2027 to Q1 2028	13,008.52	20x	388.000	670.54
Q3 2027	Q3 2027 to Q2 2028	14,116.78	20x	388.000	727.67
Q4 2027	Q4 2027 to Q3 2028	15,278.87	20x	388.000	787.57
Q1 2028	Q1 2028 to Q4 2028	16,342.59	20x	388.000	842.40
Q2 2028	Q2 2028 to Q1 2029	17,601.34	20x	388.000	907.29
Q3 2028	Q3 2028 to Q2 2029	18,831.54	20x	388.000	970.70
Q4 2028	Q4 2028 to Q3 2029	19,905.59	20x	388.000	1,026.06
Q1 2029	Q1 2029 to Q4 2029	20,735.10	20x	388.000	1,068.82
Q2 2029	Q2 2029 to Q1 2030	21,683.30	20x	388.000	1,117.70
Q3 2029	Q3 2029 to Q2 2030	22,637.64	20x	388.000	1,166.89
Q4 2029	Q4 2029 to Q3 2030	23,663.76	20x	388.000	1,219.78
Q1 2030	Q1 2030 to Q4 2030	24,753.43	20x	388.000	1,275.95

Recognition timing by ceiling

Ceiling quarter	Key inflection entering the forward window
Q3 2026	40TB ePMR ramp through HAMR conversion
Q4 2026	Higher 40TB mix through HAMR volume ramp
Q1 2027	HAMR first shipments through Wider acceptance
Q2 2027	HAMR conversion through Density migration
Q3 2027	HAMR volume ramp through Broader qualification
Q4 2027	Wider acceptance through HAMR-led supply
Q1 2028	Density migration through Demand / capacity limit
Q2 2028	Broader qualification through Density / retention
Q3 2028	HAMR-led supply through Density / retention
Q4 2028	Demand / capacity limit through Density / retention
Q1 2029	Density / retention through Density / retention
Q2 2029	Density / retention through Retained-data growth
Q3 2029	Density / retention through Density efficiency
Q4 2029	Density / retention through Capacity execution
Q1 2030	Retained-data growth through Demand / capacity limit

Complete derivation for every ceiling quarter

Q3 2026: OP window Q3 2026 to Q2 2027: $(1,880.50 + 2,165.27 + 2,474.14 + 2,820.13) \times 20 / 388.000 = \text{USD } 481.45$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2026: OP window Q4 2026 to Q3 2027: $(2,165.27 + 2,474.14 + 2,820.13 + 3,099.85) \times 20 / 388.000 = \text{USD } 544.30$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2027: OP window Q1 2027 to Q4 2027: $(2,474.14 + 2,820.13 + 3,099.85 + 3,432.33) \times 20 / 388.000 = \text{USD } 609.61$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2027: OP window Q2 2027 to Q1 2028: $(2,820.13 + 3,099.85 + 3,432.33 + 3,656.22) \times 20 / 388.000 = \text{USD } 670.54$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2027: OP window Q3 2027 to Q2 2028: $(3,099.85 + 3,432.33 + 3,656.22 + 3,928.38) \times 20 / 388.000 = \text{USD } 727.67$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2027: OP window Q4 2027 to Q3 2028: $(3,432.33 + 3,656.22 + 3,928.38 + 4,261.94) \times 20 / 388.000 = \text{USD } 787.57$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2028: OP window Q1 2028 to Q4 2028: $(3,656.22 + 3,928.38 + 4,261.94 + 4,496.05) \times 20 / 388.000 = \text{USD } 842.40$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2028: OP window Q2 2028 to Q1 2029: $(3,928.38 + 4,261.94 + 4,496.05 + 4,914.96) \times 20 / 388.000 = \text{USD } 907.29$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2028: OP window Q3 2028 to Q2 2029: $(4,261.94 + 4,496.05 + 4,914.96 + 5,158.58) \times 20 / 388.000 = \text{USD } 970.70$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2028: OP window Q4 2028 to Q3 2029: $(4,496.05 + 4,914.96 + 5,158.58 + 5,335.99) \times 20 / 388.000 = \text{USD } 1,026.06$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2029: OP window Q1 2029 to Q4 2029: $(4,914.96 + 5,158.58 + 5,335.99 + 5,325.57) \times 20 / 388.000 = \text{USD } 1,068.82$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q2 2029: OP window Q2 2029 to Q1 2030: $(5,158.58 + 5,335.99 + 5,325.57 + 5,863.16) \times 20 / 388.000 = \text{USD } 1,117.70$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q3 2029: OP window Q3 2029 to Q2 2030: $(5,335.99 + 5,325.57 + 5,863.16 + 6,112.92) \times 20 / 388.000 = \text{USD } 1,166.89$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q4 2029: OP window Q4 2029 to Q3 2030: $(5,325.57 + 5,863.16 + 6,112.92 + 6,362.11) \times 20 / 388.000 = \text{USD } 1,219.78$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

Q1 2030: OP window Q1 2030 to Q4 2030: $(5,863.16 + 6,112.92 + 6,362.11 + 6,415.24) \times 20 / 388.000 = \text{USD } 1,275.95$ per share. Each OP above is the corresponding vector GP total plus any disclosed GP adjustment, less R&D, SG&A and other opex.

The final three operating quarters support earlier four-quarter windows and have no standalone ceiling. Arithmetic uses unrounded inputs; displayed components are rounded for readability.

The denominator is 388M diluted shares in every quarter, anchored to the FQ1 FY2027 guide, versus the dated August 7 basic share count of 360.541M in the header. The difference reflects the company diluted methodology and convertible/award effects; basic and diluted counts are not interchangeable. November note settlement, employee awards and repurchases must be reconciled against later filings. Potential repurchases are assumed only to offset dilution. No future stock price is assumed to generate an artificial share-count decline.

Mechanical share sensitivity

Q1 2028 denominator	Ceiling/share
Current model	842.40
5% more shares	802.29
10% more shares	765.82

6. Cross-book relationships and monitoring

WDC <-> STX: compare density, EB, net pricing, customer acceptance and manufacturing yields. Higher industry demand helps both; one supplier gaining share must be reconciled with the other supplier model.

WDC <-> SNDK / Kioxia / Samsung / SK hynix / Micron: storage tiers share AI demand. High NAND prices help HDD economics in some uses but raise DRAM/NAND input cost inside the drive. WDC receives none of the earlier NAND +23%/+20%/+15% forecast as a direct ASP multiplier. SanDisk separation means no NAND earnings are consolidated.

WDC <-> optical / compute holdings: deployments create persistent data, but common hyperscaler capex also creates correlated customer risk. Do not add full optical, compute and memory TAMs into HDD sales.

7. Scored risks and re-rate tests

Risk class	Score	Failure channel
Technology / qualification execution	-1	HAMR reliability, yields or 40TB adoption can delay shipments and increase costs.
Demand and buyer concentration	-1	Large customers can defer data-center deployment, digest inventory or negotiate the value split.
Structural competition and substitution	-1	Seagate/Toshiba density gains, cheap NAND or performance-oriented SSD migration can limit returns.
Financial, input-cost and policy exposure	-1	Higher manufacturing capex, memory inputs, convert settlement, export limits or geographic disruption can erode cash generation.

Dated triggers and counters

Checkpoint	Positive evidence	Counter / model action
1. Next quarterly release / Q4 2026	Revenue within current guide and GM near 55-56%	Below \$4.0B or 55% requires a mix/pricing/cost bridge.
2. Q1 2027	40TB ePMR around half of nearline EB; initial qualified HAMR shipments	Cut modeled HAMR conversion if qualification does not become accepted revenue.
3. Q2-Q4 2027	HAMR scales and UltraSMR approaches fiscal-year-end target	Keep overlapping technology shares distinct; missed yields reduce capacity and margin.
4. November 16, 2026 / next filing	Convertible settlement and reported diluted shares reconcile	Update the denominator only on actual or guided settlement economics.
5. 2028 and each subsequent year	Capex, EB and retained-data demand support the modeled growth	Lower supply/ASP assumptions if customer inventory or competing capacity rises; no second ceiling probability haircut.

Consolidated near-term guidance is HIGH confidence; product-level profit allocation and the cloud-only EB split are proxies, not disclosures. Beyond 2027 all operating extensions are LOW confidence. Current Yahoo high-revenue-estimate tables could not be retrieved, so that cross-check is unavailable and no stale snippet is treated as a floor. HAMR probabilities apply

once at revenue. The artifact is a single base case, not a claim that every announced long-range capacity or performance goal is delivered on time.

8. Sources and snapshot

- **WDC-1** - August 5, 2026. FY2026 Q4 release and financial reconciliation. Latest actuals, prior-quarter comparison and FQ1 FY2027 guidance.
- **WDC-2** - August 14, 2026. FY2026 Form 10-K. Standalone HDD perimeter, competitors, share snapshot and risks.
- **WDC-3** - August 5, 2026. FY2026 Q4 management call. Latest earnings-call operating milestones, end-market revenue and EB.
- **WDC-4** - April 30, 2026. FY2026 Q3 management call. Second latest call: pricing mechanics, qualification and manufacturing lead times.
- **WDC-5** - September 9, 2026. Citi conference - Western Digital management. More recent HAMR timing, cost headwinds and capex commentary.
- **WDC-6** - September 21, 2026 close. WDC dated market snapshot. Header price only; does not determine Base or ceilings.
- **WDC-7** - September 14, 2026. Convertible redemption notice - Form 8-K. November 16 planned redemption, principal outstanding and conversion terms.
- **WDC-8** - September 15, 2026. WD AI storage research update. Company-sponsored survey supports direction of data growth; not added as a separate sales forecast.
- **D1** - July 30, 2026. Amazon Q2 management call. Management cash-capex outlook: \$220B in CY2026; increase partly reflects memory cost inflation.
- **D2** - July 29, 2026. Microsoft FY2026 Q4 management call. CY2026 capex about \$175B after lease classification change; physical investment expectations unchanged.
- **D3** - July 22, 2026. Alphabet Q2 management call. CY2026 capex \$195-205B; use \$200B midpoint.
- **D4** - July 29, 2026. Meta Q2 results. CY2026 capex including principal finance-lease payments \$130-145B; use \$137.5B midpoint.

Model snapshot: September 21, 2026. WDC. Framework 6.0.2. Base 46/60 (77%). Eighteen modeled operating quarters and fifteen complete forward-four-quarter ceilings. Current full PDF and synchronized Markdown: G:/My Drive/1amInvesting/Artifacts/WDC_WesternDigital_DeepDive.pdf and .md. The companion company worksheet uses this same model and links its ceiling series to the master portfolio.